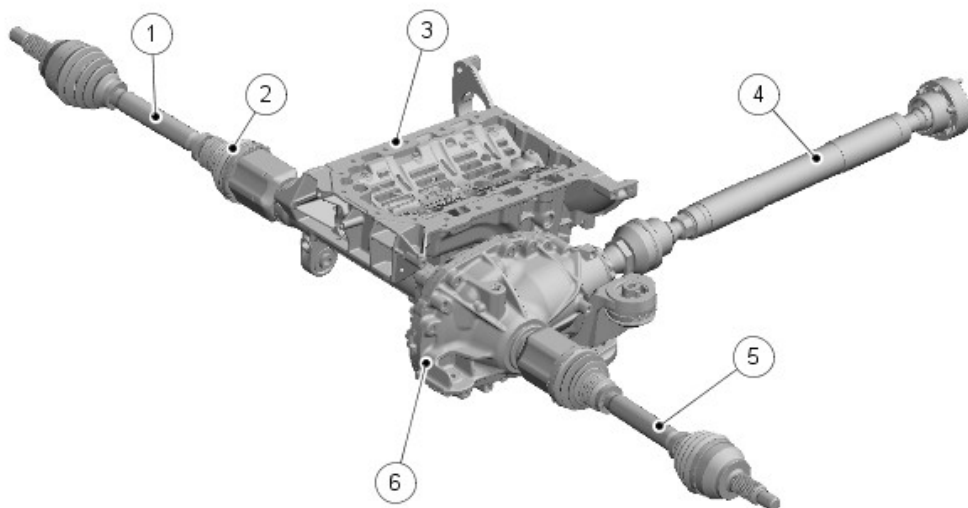


2016.0 RANGE ROVER (LG), 205-03

FRONT DRIVE AXLE/DIFFERENTIAL

DESCRIPTION AND OPERATION

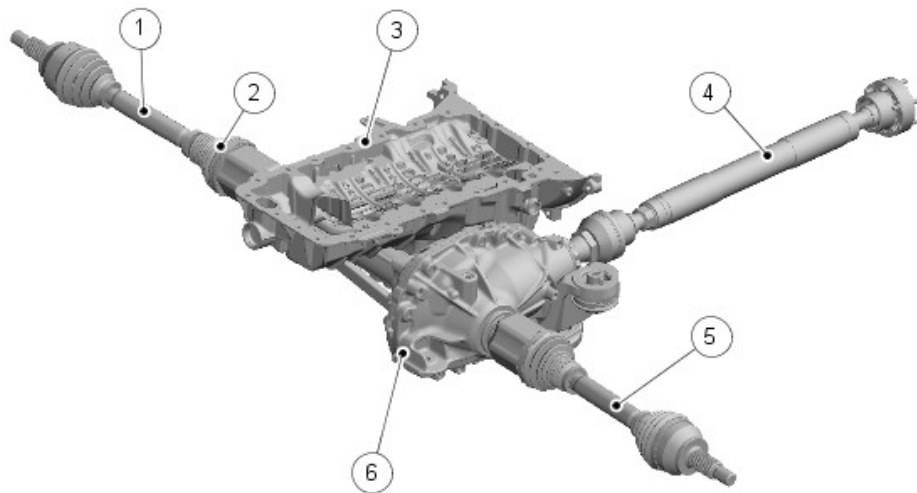
TDV6 3.0L DIESEL



E148984

ITEM	DESCRIPTION
1	Front right halfshaft
2	Constant Velocity (CV) joint
3	Engine oil pan
4	Front driveshaft
5	Front left halfshaft
6	Front differential

TDV8 4.4L DIESEL



E148985

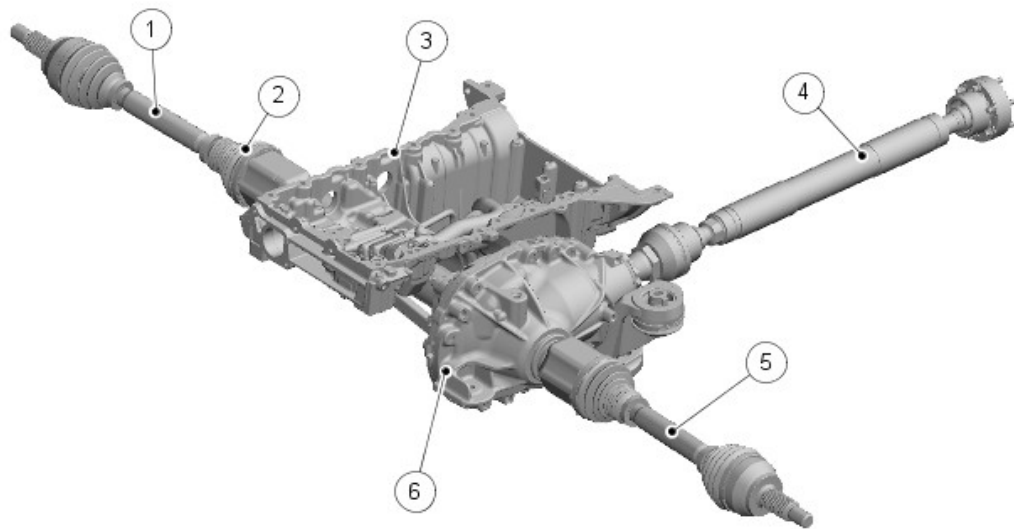
ITEM	DESCRIPTION
1	Front right halfshaft

2	CV joint
3	Engine oil pan
4	Front driveshaft
5	Front left halfshaft
6	Front differential

PETROL Engines

NOTE:

V6 3.0L S/C PETROL shown, V8 5.0L PETROL and V8 S/C 5.0L PETROL are similar.



E148983

ITEM	DESCRIPTION
1	Front right halfshaft
2	CV joint
3	Engine oil pan
4	Front driveshaft
5	Front left halfshaft
6	Front differential

OVERVIEW

The front differential converts the 'angle of drive' through 90 degrees and

distributes drive, via the halfshafts, to the front wheels.

The front differential is mounted upon the front subframe on 3 mounts. The differential has a pinion nose bracket situated at the rear left, which incorporates a bush with a limiter washer. The differential is bolted through this bush into the subframe. The front left hand front mount has the bush situated in the subframe. The differential has a tube on the right side of the differential carrier, this tube limits the torque steer.

The casing comprises two halves with machined mating faces. When assembled, the iron casing halves are sealed with a thin film of Loctite 5999 sealant and secured together with 12 bolts. The left casing is the carrier for all the rotating parts and the right casing is a cover to close the unit and a support for the right carrier bearing. A breather tube is fitted to the casings. This allows a plastic tube to be fitted and routed to a high point in the engine compartment, preventing the ingress of water when the vehicle is wading.

The differential is a conventional design using a hypoid gear layout. This employs a hypoid bevel pinion gear and crown wheel, with the pinion offset above the center line of the crown wheel. This design allows for a larger pinion gear to be used, which has the advantages of increased gear strength and reduced operating noise.

The output ratio of the front differential varies depending on the engine variant.

For additional information, refer to: Specifications (205-03 Front Drive Axle/Differential, Specifications).

The ratio is changed by changing the amount of teeth between the crown wheel drive gear and pinion gear. For example, with a ratio of 3.21:1, the crown wheel drive gear will have 3.21 times more teeth than the pinion gear.

The differential comprises a pinion shaft and hypoid bevel gear, a crown wheel drive gear with an integral cage, which houses two planet gears. Two

sun wheels are also located in the cage and pass the rotational drive to the drive halfshaft shafts.

The spacer is used to hold the bearings in alignment and collapses at a predetermined rate due the torque applied to the pinion nut. The remaining load is carried as preload in the bearings and is set with the pinion nut to give the desired assembly torque-to-turn. This can only be performed during first assembly when new bearings are coated with friction modifier.

An oil seal is pressed into the left casing and seals the input flange to the differential unit. The pinion shaft has a hypoid bevel gear at its inner end, which mates with the crown wheel drive gear.

The crown wheel drive gear is located on the carrier and secured with 10 screws. The carrier is mounted on taper roller bearings located in each casing half. The bearings are press fitted into the casing and a spacer is located on the outside face to apply preload to the bearing.

The carrier is fitted with a shaft onto which the two planet gears are mounted. The shaft is secured in the carrier with a roll pin. The sun wheels are located in pockets within the carrier and mesh with the planet gears. Curved plates are located between the carrier and the sun wheels and hold the sun wheels in mesh with the planet gears. Each sun wheel has a machined, splined, bore to accept the drive halfshaft. A groove is machined in the bore to locate the circlip fitted to the drive halfshaft, providing positive drive halfshaft location.

The operating principles of the front and rear differential are the same. For additional information, refer to: Rear Drive Axle and Differential (205-02 Rear Drive Axle/Differential, Description and Operation).